CASCADIA ANALYTICS

ASML Holding N.V.

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Guidance Step-Ups Collide with Multiple Compression Risk

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ASML continues to demonstrate operational execution across advanced semiconductor lithography, highlighted by three successive guidance increases through FY2026 and quarterly gross margins reaching 57.5% in Q3 FY2026. However, market pricing at 68.0x latest fiscal year diluted earnings and an enterprise value multiple of 51.4x EBITDA leaves no margin for shipment timing friction or broader valuation multiple mean reversion. While system demand remains intense and the backlog provides operational visibility, the risk-reward profile is skewed negatively at prevailing spot levels. Our probability-weighted discounted cash flow framework yields a blended target of $1,450.27, representing -13.7% downside versus the reference spot price of $1,680.73.

HOLD	$1,450.27	-13.7%	$1,680.73
Hold, Valuation Concerns	BLENDED TARGET	BLENDED UPSIDE	REFERENCE PRICE
INVESTMENT RATING			

The market remains fixated on record quarterly prints and headline guidance step-ups to €44B to €45B, yet it systematically overlooks the heightened valuation vulnerability embedded in an enterprise trading at 20.0x latest fiscal year revenue. The reality is that even exceptional structural moats cannot insulate long-duration

cash flows from multiple normalization when baseline expectations fully price flawless operational execution.

Executive summary

ASML operates as the monopoly supplier of extreme ultraviolet (EUV) lithography systems for advanced semiconductor manufacturing globally. The company generates revenue through advanced system sales, spanning EUV systems, deep ultraviolet (DUV) platforms, and metrology solutions, paired with a high-margin Installed Base Management business that delivers critical services and field upgrades. This commercial positioning makes the franchise an essential foundation for leading-edge silicon fabrication, creating extraordinary customer lock-in and high barriers to entry that have historically supported superior financial returns.

Recent operational prints confirm that operational momentum accelerated through the second half of FY2026. In Q3 FY2026, ASML generated record quarterly net sales of €11.66B, representing 55.2% year-over-year expansion, accompanied by a gross margin of 57.5% and Installed Base Management revenue of €3.29B, which exceeded management guidance. This performance allowed leadership to raise full-year 2026 net sales guidance to a range of €44B to €45B, marking the third consecutive guidance increase of the fiscal year, with Q4 net sales guided to €14.2B to €15.2B. However, the print landed below the roughly €11.85B market consensus, and net income of €4.96B included a non-operating revaluation gain on the Mistral AI investment, which overstates underlying operating earnings.

From a balance sheet perspective, the business maintains a fortress liquidity architecture. As of the close of FY2025, ASML reported cash of €12.9B against total debt of €4.39B, yielding a net cash position of €8.53B. The company generated €11.0B in free cash flow on revenue of €32.7B in FY2025, producing a free cash flow margin of 33.8%, an operating margin of 34.6%, and return on equity of 49.0%. Capital return policies have remained shareholder-oriented, reinforced by the announcement of a share buyback program of up to €12B alongside the Q4 FY2025 results.

Notwithstanding this elite fundamental profile, current market valuations fully reflect these strengths. At the reference spot price of $1,680.73, ASML carries a market capitalisation of €653.6B, trading at 68.0x latest fiscal year diluted earnings of $24.71 and an EV to revenue multiple of 19.7x. With a modeled weighted average cost of capital of 9.80% and equity beta of 1.30, our DCF framework points to significant valuation compression across base and bear scenarios. Consequently, we initiate coverage with a rating of Hold, Valuation Concerns, anchored to our blended price target of $1,450.27.

Operational snapshot

$1,680.73	€32.7B	€11.0B	33.8%	1.30	$1,680.73
REF. PRICE	FY2025 REVENUE	FY2025 FREE CASH FLOW	FCF MARGIN	EQUITY BETA	REF. CLOSE

Fiscal-year figures as reported in company filings and collected by Cascadia Analytics; ref. price as of 2026-10-29.

The moat is Absolute and Monopolistic

ASML's competitive advantage rests upon an insurmountable intellectual property, optical engineering, and supply chain ecosystem. As the sole supplier of EUV lithography systems globally, the company commands an absolute monopoly in the sub-nanometer patterning tools essential for high-volume manufacturing of cutting-edge logic and memory devices. Customers seeking to build leading-edge semiconductor nodes have no alternative manufacturing pathway, granting ASML unmatched pricing power that is clearly evident in gross margins expanding from 52.8% in FY2025 to 57.5% in Q3 FY2026.

The switching costs associated with ASML's lithography platforms are effectively prohibitive. Semiconductor fabricators design entire multi-billion-dollar cleanroom architectures, chemical resist processes, and mask tooling standards around ASML's EUV, DUV, and metrology systems. Replacing or altering these production lines is economically unviable, ensuring structural customer retention across entire technological generations. This deep integration is reinforced by the Installed Base Management segment, which generated €3.29B in Q3 FY2026, as clients continually invest in service contracts, software optimizations, and hardware upgrades to sustain fab uptime.

Furthermore, the barrier to market entry is fortified by decades of specialized co-development with key supply partners and research consortiums. Replicating the complex light sources, mirrors, vacuum systems, and precision stages required for EUV lithography would require billions of euros in speculative capital and decades of trial-and-error engineering without guaranteed success. Even in mature DUV lithography and metrology, ASML retains immense customer entrenchment, ensuring that fab expansions across both mature and advanced trailing nodes remain captive revenue sources.

Valuation

Three scenarios make the valuation range explicit

Our valuation framework implements a three-scenario discounted cash flow model calibrated over a forecast horizon running from FY2026 through FY2031, discounting unlevered free cash flows at a verified WACC of 9.80% with 388.9 million diluted shares outstanding. Baseline cash flow generation builds upon the verified FY2025 financial base, which established revenue of

€32.7B, operating income of €11.3B, and free cash flow of €11.0B at a 33.8% free cash flow margin. Terminal value is determined using a Gordon growth methodology applying a 3.0% terminal growth rate in the base case.

The model delineates fundamental trajectories across three distinct operating paths. In the Bear Case, top-line revenue compounds at a 9.0% CAGR through FY2031, with final forecast year free cash flow margins normalizing to 33.2%, producing a target valuation of $1,113.73, representing -33.7% downside versus spot. In the Base Case, revenue compounds at an 11.5% CAGR with a terminal free cash flow margin reaching 35.6%, yielding a target price of $1,410.46, or -16.1% downside. In the Bull Case, sustained high-volume system shipments drive a 14.6% revenue CAGR with final forecast year free cash flow margins expanding to 37.9%, generating a fair value of $1,866.43, representing +11.0% upside.

Weighting these outcomes at 25% for the Bear Case, 50% for the Base Case, and 25% for the Bull Case generates a blended target price of $1,450.27, representing a -13.7% return potential relative to the reference spot price of $1,680.73. At prevailing market quotes, the equity market is pricing in cash flow duration and operating margins that exceed even our Bull Case assumptions, leaving current market pricing heavily exposed to negative revisions.

SCENARIO	WEIGHT	REVENUE CAGR	TARGET FCF MARGIN	TARGET	UPSIDE
Bear	25%	9.0%	33.2%	$1,113.73	-33.7%
Base	50%	11.5%	35.6%	$1,410.46	-16.1%
Bull	25%	14.6%	37.9%	$1,866.43	+11.0%

Bear $1,113.73

Reflects shipment pacing bottlenecks, a deceleration in trailing-node spending, and multiple compression, driving a 9.0% revenue CAGR and a target FCF margin of 33.2%. Incorporates the risk that quarterly consensus misses, such as that seen in Q3 FY2026 net sales, become recurring as expectations outstrip operational delivery. Yields an equity value representing -33.7%

Base $1,410.46

Models steady execution across the system backlog and robust demand for Installed Base Management, resulting in an 11.5% revenue CAGR and a 35.6% target FCF margin. Reflects the raised FY2026 guidance range of €44B to €45B while acknowledging the long-term reality of cyclical digestion in semiconductor capex. Delivers a fair value representing -16.1% downside from spot.

Bull $1,866.43

Assumes accelerated customer fab buildouts and premium pricing across EUV and metrology platforms, yielding a 14.6% revenue CAGR and a 37.9% target FCF margin. Driven by continuous guidance increases and rapid conversion of the €38.80B backlog, generating +11.0% upside versus the reference price.

downside versus the reference spot price.

Valuation framework

A live discount rate, not a static hurdle

Our valuation framework evaluates ASML's cash flows under a dynamic 9.80% weighted average cost of capital, grounded in an equity beta of 1.30. In our assessment, this discount rate accurately captures the balance between ASML's absolute operational monopoly in EUV lithography and the systematic macro sensitivity inherent in the capital expenditure cycles of global semiconductor manufacturers. While the business carries an exceptionally low financial risk profile with net cash of €8.53B and total debt of only €4.39B, its operational cash flows remain tethered to the timing of multi-billion-dollar fabrication facility openings.

The 9.80% discount hurdle reflects the reality that capital deployed into high-multiple semiconductor equipment equities requires an adequate risk premium against long-duration interest rate uncertainty. At a reference spot price of $1,680.73 and a market capitalization of €653.6B, the prevailing valuation implies that the equity market is discounting terminal cash flows at a rate well below 9.80%, effectively mispricing the cyclicality of fab equipment delivery schedules and the reinvestment hurdles required to sustain high-volume manufacturing tool availability.

FCF margin normalization is the engine

Free cash flow margin trajectory serves as the primary quantitative bridge across our three scenario outcomes. In FY2025, ASML converted €32.7B in net revenue into €11.0B in free cash flow, representing a robust FCF margin of 33.8%, supported by an operating margin of 34.6% and a gross margin of 52.8%. Over recent quarters, operating profitability has demonstrated significant operating leverage, with operating margins tracking at 36.0% in Q1 FY2026, 37.1% in Q2 FY2026, and gross margins surging to 57.5% in Q3 FY2026, bolstered by high-margin Installed Base Management revenue of €3.29B.

Our model incorporates structural margin normalization as system complexity and customer delivery schedules expand across the FY2026 to FY2031 forecast period. In the Base

Case, the terminal FCF margin stabilizes at 35.6%, benefiting from the expanding installed tool base and recurring service contracts. In the Bull Case, premium platform realizations expand the target FCF margin to 37.9%, whereas the Bear Case models margin compression to 33.2% as input costs, field support requirements, and cleanroom integration investments dilute free cash flow conversion. Cash flow generation across all paths remains disciplined, but the valuation outcome is acutely sensitive to where conversion settles within this 33.2% to 37.9% band.

Mechanics are explicit

Revenue becomes EBIT, cash taxes come out, then we add back D&A, subtract CapEx and the change in net working capital to reach unlevered free cash flow, which we discount and divide by diluted shares.

> **Methodology.** Detailed EBIT-bridge DCF on three scenarios, probability-weighted 25/50/25. Beta decays toward a mature peer over the forecast. Terminal value via Gordon growth. Targets are model-computed, not estimated.

KEY VALUATION PARAMETERS

Forecast window	**FY2026 to FY2031**
Live WACC	**9.80% (2026-10-29)**
Terminal growth	**3.0%**
Equity beta	**1.30**
Diluted shares	**388.9 million**

HOLD

over $1,450.27 blended target

Competitive moat and concentration risk

We rate the moat as Wide and we view the market structure as Monopoly (EUV lithography). We assign a premium of +35% to sector median multiples on moat strength alone.

- sole-supplier EUV + High NA, first High NA shipments 2023, Intel past 1m High NA wafers, Samsung and SK Hynix adopting High NA from 2028 (Reuters, Sep 2026).
- FY2025 net sales €32.7bn (+15% YoY), EPS €24.73 (+28.5%), 2026 guidance €34-39bn at 51-53% gross margin, year-end 2025 backlog €38.8bn (company/press coverage, 2026); SK Hynix disclosed a ~$8bn EUV order on 24 Mar (delivery by 31 Dec 2027, ~30 tools at $250-300m each, Bernstein estimate). A €38.8bn backlog at 51-53% guided gross margin is durable pricing power, not a spot spike.
- The weakest element is geopolitical/regulatory exposure, export controls on China sales and single-customer concentration (TSMC, Samsung, Intel, SK Hynix, Micron) cap the rating below an unconditional premium, but do not remove Wide, no rival can replicate the installed base, service network, or Zeiss optics chain within a planning horizon.

On a multiple cross-check, the justified range is 24.0-27.0x / 20.9-23.5x. The cross-check value is $1,313-$1,477 vs Base $1,203.94, spot $1,603.91, on a base of Framework base from the October 27 moat report. The gap verdict is In part, materially (~68% of gap at top); spot P/E 25.5x still above justified ceiling.

Backlog is €38.8bn system backlog, 31 Dec 2025, with coverage of 14.2 months vs FY2025 sales, 10.5 months vs 2026 guide and visibility rated High. On backlog alone, the stance is Supports.

We score concentration and credit risk across seven families, where a High score is 8 points, a Medium score is 4 points, and a Low score is 1 point.

Risk family	Score
customer concentration	8 points
counterparty credit and funding dependence	1 points
single product, single program, or single platform dependency	4 points
single supplier, single fab, or single geography exposure	8 points
platform, channel, or distribution dependency	1 points
customer insourcing risk	1 points
regulatory single point risk	8 points

The combined discount is 20 percent, capped at 20 percent, against an uncapped total of 31 percent. Shown side by side with the moat premium or discount, the net stance is +15 pp points arithmetically. We do not net these figures into a new target, and we do not double count a family excluded under the double-count rule.

The ladder ranks risk coarsely, and because 32 of the 47 names reach the 20 percent cap, the combined discount is not a mechanical valuation haircut until the ladder is recalibrated.

On the gap to spot, the concentration assessment reads as follows. Not at all

Both the moat premium or discount and the concentration discount are multiple and risk cross-checks dated October 27, 2026, and neither figure is in the DCF target itself. Cross-check figures were computed against the Base in the October 27 framework, so they may differ from a live Base that has since moved.

Business & segment drivers

Revenue prospects anchor on Advanced System Shipments and Installed Base Expansion HOLD

ASML's top-line architecture is structured around two distinct operational pillars, advanced lithography system sales and Installed Base Management. System revenue encompasses extreme ultraviolet systems, deep ultraviolet immersion and dry tools, and precision metrology

and inspection systems. EUV systems remain the defining technological vector, as semiconductor manufacturers require these tools for advanced node transitions. Demand visibility was strongly reinforced in Q4 FY2025, when ASML recorded net bookings of €13.16B, including €7.44B in EUV orders, expanding the total order backlog to €38.80B.

The second pillar, Installed Base Management, delivers high-margin recurring cash flow through maintenance, system enhancements, and field upgrades across the operating fleet. In Q3 FY2026, Installed Base Management net sales reached €3.29B, outperforming management's guidance of approximately €2.9B. This recurring revenue stream acts as an operational stabilizer, decoupling a portion of ASML's cash flow from quarterly new tool installation schedules while reinforcing fab-level customer retention.

Delivery execution accelerated across FY2026 following a measured opening quarter. Net sales advanced from €8.77B in Q1 FY2026, up 13.2% year-over-year, to €9.33B in Q2 FY2026, representing 21.3% growth, before surging to record net sales of €11.66B in Q3 FY2026, up 55.2% year-over-year. This cadence justified management raising full-year 2026 sales guidance from its initial January range of €34B to €39B up to €44B to €45B by October, with Q4 FY2026 net sales guided to a steep ramp of €14.2B to €15.2B.

Despite this operational progression, market expectations have escalated to levels that penalize even minor variance. In Q3 FY2026, despite record net sales and a 57.5% gross margin, the print landed below the roughly €11.85B market consensus, causing the stock to gain only 0.5% in the post-earnings session. Furthermore, reported net income of €4.96B was elevated by a revaluation gain on the Mistral AI investment, which produced basic EPS of €12.95 against a dashboard comparison figure of €11.82 and a consensus estimate of €10.76, obscuring the normalized underlying earnings run rate.

What to monitor

- **Guidance execution in Q4 FY2026** Tracking net sales against the ambitious guided target of €14.2B to €15.2B required to deliver full-year revenue of €44B to €45B.
- **Installed Base Management run rate** Monitoring service and upgrade net sales against the €3.29B level established in Q3 FY2026 to assess recurring margin durability.
- **Backlog replenishment dynamics** Observing whether quarterly net bookings match the €13.16B milestone set in Q4 FY2025 to sustain visibility past the €38.80B backlog level.
- **Gross margin stability** Evaluating whether gross margins can hold near the 57.5% mark achieved in Q3 FY2026 as product mix evolves across EUV and DUV shipments.

+15.6%	52.8%	34.6%	33.8%	€8.53B	49.0%
FY2025 REVENUE GROWTH	GROSS MARGIN	OPERATING MARGIN	FCF MARGIN	NET CASH	RETURN ON EQUITY

METRIC	FY2025 (AS REPORTED)
Revenue	€32.7B
Gross Profit	€17.3B
EBIT	€11.3B
Net Income	€9.61B
EBITDA	€12.6B
Free Cash Flow	€11.0B
Diluted EPS	$24.71
ROE	49.0%

Risk & moat matrix

The bear case

SHARE EROSION

While EUV remains an absolute monopoly, customer diversification into alternative manufacturing processes could limit long-term tool intensity. Any slowing in leading-edge transition cadences would directly constrain total addressable volume.

M&A DIGESTION

Non-operating balance sheet investments introduce earnings noise and accounting volatility, as shown in Q3 FY2026 when net income included a Mistral AI revaluation gain. Such gains distort underlying operating metrics and elevate headline

REGULATORY AND PRICING SCRUTINY

Operating as a sole-source monopoly exposes system sales to heightened geopolitical trade controls and customer pushback against equipment pricing. Export restrictions could impede the fulfillment of advanced DUV and EUV system deliveries to key regional markets.

TECHNOLOGY DISRUPTION

Customer delays in adopting next-generation manufacturing platforms could elongate tool delivery schedules and cause backlog slippage. A mismatch between fab construction readiness and tool availability risks deferring revenue recognition across quarters.

REINVESTMENT TREADMILL

Sustaining lithography leadership demands continuous capital deployment, which could compress free cash flow conversion if execution costs rise. Failure to maintain operational leverage would pull cash flow conversion down toward our Bear Case margin of 33.2%.

WACC AND DURATION SENSITIVITY

With an equity beta of 1.30 and a 9.80% WACC, ASML's valuation is highly vulnerable to interest rate shifts at 68.0x latest fiscal year earnings. Any upward pressure on discount rates disproportionately depresses the present value of distant forecast cash flows.

basic EPS to €12.95.

The strengths

OLIGOPOLY POSITION

ASML holds an unchallenged global monopoly in EUV lithography systems, rendering advanced chipmakers wholly reliant on its product suite. This gives the company pricing leverage and defensible operating margins across all major market environments.

FCF COMPOUNDING

The company generated €11.0B in free cash flow in FY2025 on a 33.8% FCF margin, supported by an operating margin of 34.6%. High return on equity of 49.0% provides ample organic liquidity to support capacity growth alongside the €12B buyback program.

GEOGRAPHIC DURABILITY

A global installed base ensures that high-margin Installed Base Management revenue compounds steadily regardless of single-region volatility. Installed base net sales of €3.29B in Q3 FY2026 showcase the resilience of service agreements across the worldwide fleet.

PIPELINE OPTIONALITY

A robust backlog of €38.80B, built upon record quarterly bookings of €13.16B in Q4 FY2025, provides deep multi-year revenue visibility. This order visibility anchors management's raised FY2026 revenue guidance of €44B to €45B.

The bottom line on risk and reward

ASML remains an exceptional franchise that operates with unmatched technological power and monopolistic industry defensibility. The cadence of FY2026 earnings has demonstrated clear operating leverage, transitioning from €8.77B in Q1 net sales to €11.66B in Q3, accompanied by gross margins climbing to 57.5%. The commercial viability of the business is supported by €8.53B in net cash, an expanding service revenue base, and a €38.80B backlog that shields multi-year operational planning from short-term customer fluctuations.

However, from an equity investment perspective, operational excellence must be weighed against prevailing valuation multiples. At $1,680.73, the market capitalisation of €653.6B prices the stock at 20.0x latest fiscal year sales and 51.4x EBITDA, multiples that leave virtually zero tolerance for execution slips, fab delivery adjustments, or general valuation normalization. The restrained share price reaction of +0.5% following the Q3 FY2026 print confirms that the market requires increasingly flawless quarterly beats to sustain valuation expansion.

Our probability-weighted DCF model establishes a blended target price of $1,450.27, which indicates a downside of -13.7% relative to the current reference spot price. With the Base Case yielding $1,410.46 (-16.1%) and the Bear Case indicating $1,113.73 (-33.7%), the prospective risk-reward balance remains unfavorable for capital deployment. We advise

institutional investors to maintain a Hold rating until market multiples compress to levels that better align with sustainable long-term cash flow compounding.

Justin Wen, Senior Equity Research Analyst, Cascadia Analytics

Franchise deep dive

Monopoly EUV Lithography and Installed Base Management: Structural Pillars of Advanced Fabrication

ASML's franchise strength is anchored in its monopoly position as the sole provider of extreme ultraviolet lithography systems globally, complemented by deep ultraviolet systems, precision metrology, and high-margin Installed Base Management services. Lithography tools represent the single most capital-intensive and technologically critical phase of semiconductor fabrication, placing ASML at the center of all leading-edge microchip manufacturing roadmaps. Without ASML's EUV platforms, fabricating advanced logic chips and high-density memory at commercial scale is physically impossible under current industry architectures.

The company's commercial model combines multi-million-euro hardware sales with high-margin recurring operational revenue. System revenue, which incorporates EUV, DUV, and metrology equipment, creates the operational baseline for fab cleanrooms worldwide. Once tools are installed, ASML captures long-tail economic value through Installed Base Management, providing technical maintenance, component upgrades, and yield-enhancing enhancements that run for decades. In Q3 FY2026, Installed Base Management net sales reached €3.29B, surpassing the guided level of roughly €2.9B and playing a central role in lifting gross margins to 57.5%.

The backlog visibility underpinning this franchise is historically exceptional. At the conclusion of Q4 FY2025, net bookings achieved a record quarterly level of €13.16B, including €7.44B specifically for EUV systems, elevating the total order backlog to €38.80B. This monumental order book allowed management to initiate FY2026 net sales guidance at €34B to €39B, which was subsequently raised in successive quarters to €44B to €45B by Q3 FY2026. However, fulfilling these backlogs requires extensive operational coordination, making the business sensitive to fab build schedules and second-half delivery ramps.

What the recent record shows

- Record quarterly net bookings of €13.16B in Q4 FY2025, including €7.44B of EUV orders, which expanded the total backlog to €38.80B.
- Successive full-year 2026 revenue guidance increases, moving from an initial outlook of €34B to €39B up to €44B to €45B by Q3 FY2026.
- Accelerating quarterly net sales throughout FY2026, advancing from €8.77B in Q1 to €9.33B in Q2, and reaching a record €11.66B in Q3.

- Installed Base Management net sales reaching €3.29B in Q3 FY2026, beating guided levels of approximately €2.9B and driving gross margin to 57.5%.

> **Our read:** ASML's lithography monopoly and growing service revenues confirm its status as an elite technology franchise. However, with the stock trading at $1,680.73, market pricing has outrun fundamentals, pointing to our blended target of $1,450.27 and warranting our Hold, Valuation Concerns stance.